CHAPTER 5 PROJECT

IoT Based Smart Helmet for Alcohol Detection, Accident Monitoring, and Real-Time GPS Alerts

1.1 Objective

The objective of this project is to design and implement an IoT-based smart helmet that detects rider alcohol consumption, monitors accidents in real time, and sends GPS-based emergency alerts to predefined contacts. The system enhances two-wheeler safety by preventing vehicle ignition when alcohol is detected and by automatically notifying guardians or emergency services in case of a crash.

1.2 Scope of the Project

- Designing a smart helmet that integrates alcohol detection, accident detection, and GPS/GSM-based alerting in a single embedded system.
- Implementing ignition control logic so that the vehicle starts only when the rider is sober and the helmet is worn.
- Continuously monitoring motion and impact using sensors to detect accidents and trigger automatic emergency alerts with live location.
- Developing and testing firmware on a microcontroller platform (e.g., Arduino, ESP32) for sensor interfacing, decision logic, and communication.
- Validating system performance in terms of detection accuracy, response time, reliability of alerts, and practical usability.

2. System Overview

An IoT-based smart helmet integrates sensing, processing, and communication modules to improve rider safety and enable real-time monitoring. The system continuously checks alcohol levels, helmet usage, and impact events, and then uses GSM/GPS modules to send alerts with coordinates when a critical condition is detected.

2.1 Core Functions to Implement

- Alcohol Detection:
    - Breath alcohol sensing near the rider’s mouth using an MQ3 or equivalent alcohol sensor.

    - Threshold-based decision (e.g., alcohol concentration above a preset limit) to block vehicle ignition and trigger warnings.
- Helmet Wearing Detection:
    - Ensuring the engine can start only when the helmet is properly worn using IR/proximity or pressure sensors.
- Accident Detection:
    - Using an accelerometer/gyroscope (e.g., MPU6050) or vibration sensor (SW-420) to detect abnormal acceleration or impacts.
    - Confirming accident events based on configurable thresholds to minimize false positives.
- Real-Time GPS Tracking & Alerts:
    - Acquiring latitude and longitude via a GPS module (e.g., NEO-6M).
    - Sending SMS alerts with location links to registered contacts through a GSM module (e.g., SIM800L).
- Ignition Control & Safety Logic:
    - Controlling bike ignition via relay based on alcohol level and helmet-wear status.
    - Providing buzzer/LED indications for different system states (safe, alcohol detected, accident detected, alert sent).

3. Hardware Design

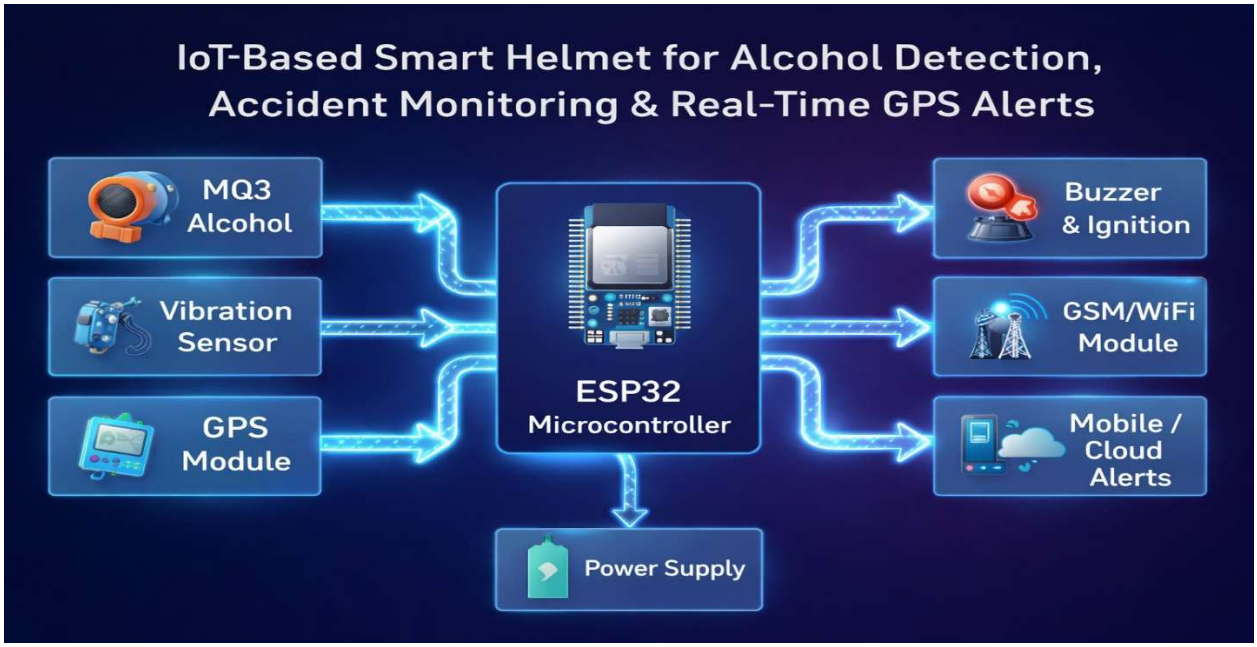
3.1 Major Components

- Microcontroller board (Arduino Uno / ESP32) for sensor interfacing, decision making, and communication.
- MQ3 alcohol sensor for detecting alcohol vapors in the rider's breath.
- Accelerometer/gyroscope module (MPU6050) or vibration sensor (SW-420) for accident detection.
- GPS module (NEO-6M) for real-time location tracking.
- GSM module (SIM800L or similar) for SMS-based emergency alerts.

- Helmet-wear detection sensor (IR proximity sensor / pressure switch) to confirm helmet usage.
- Relay module to enable/disable vehicle ignition based on safety logic.
- Power supply (battery, voltage regulators) and supporting components (wires, connectors, PCB/breadboard).

3.2 Block Diagram Description

The helmet's sensors (alcohol, impact, wear-detection) feed data to the microcontroller, which processes inputs and triggers outputs such as ignition relay, buzzer, and communication modules. The GPS module provides coordinates, which are combined with preformatted alert messages and sent via GSM to emergency contacts in case of accidents or high alcohol levels.

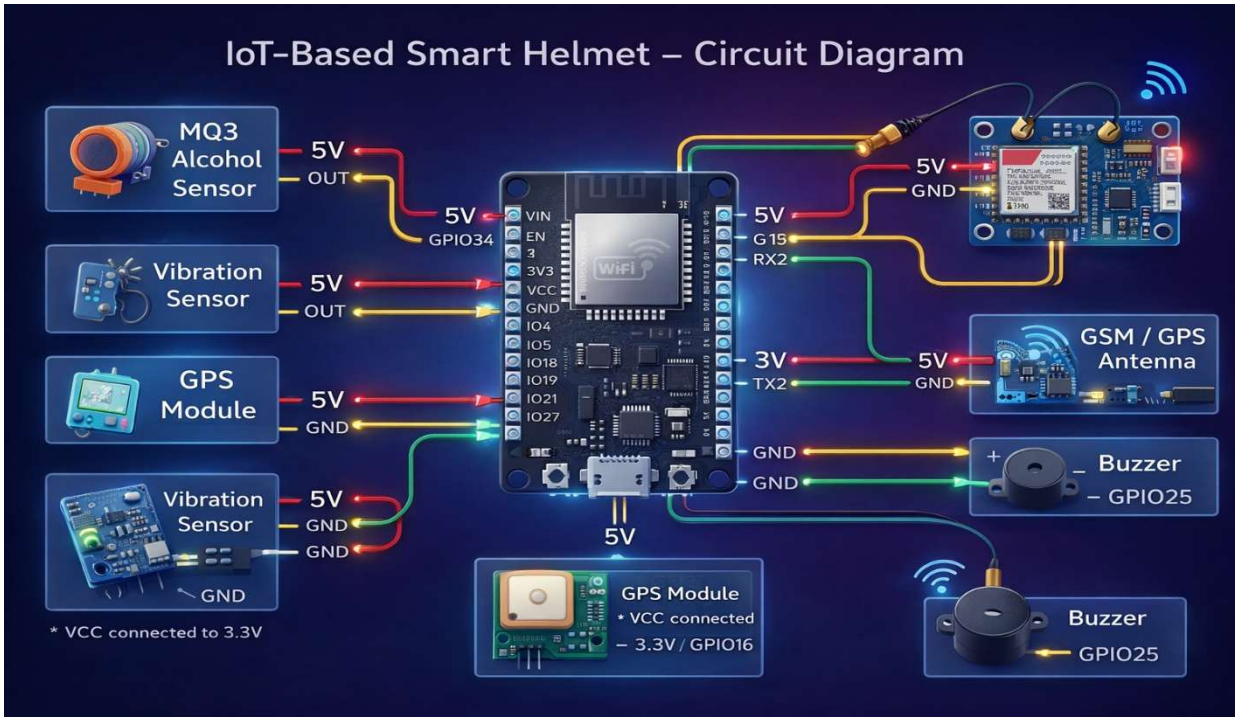


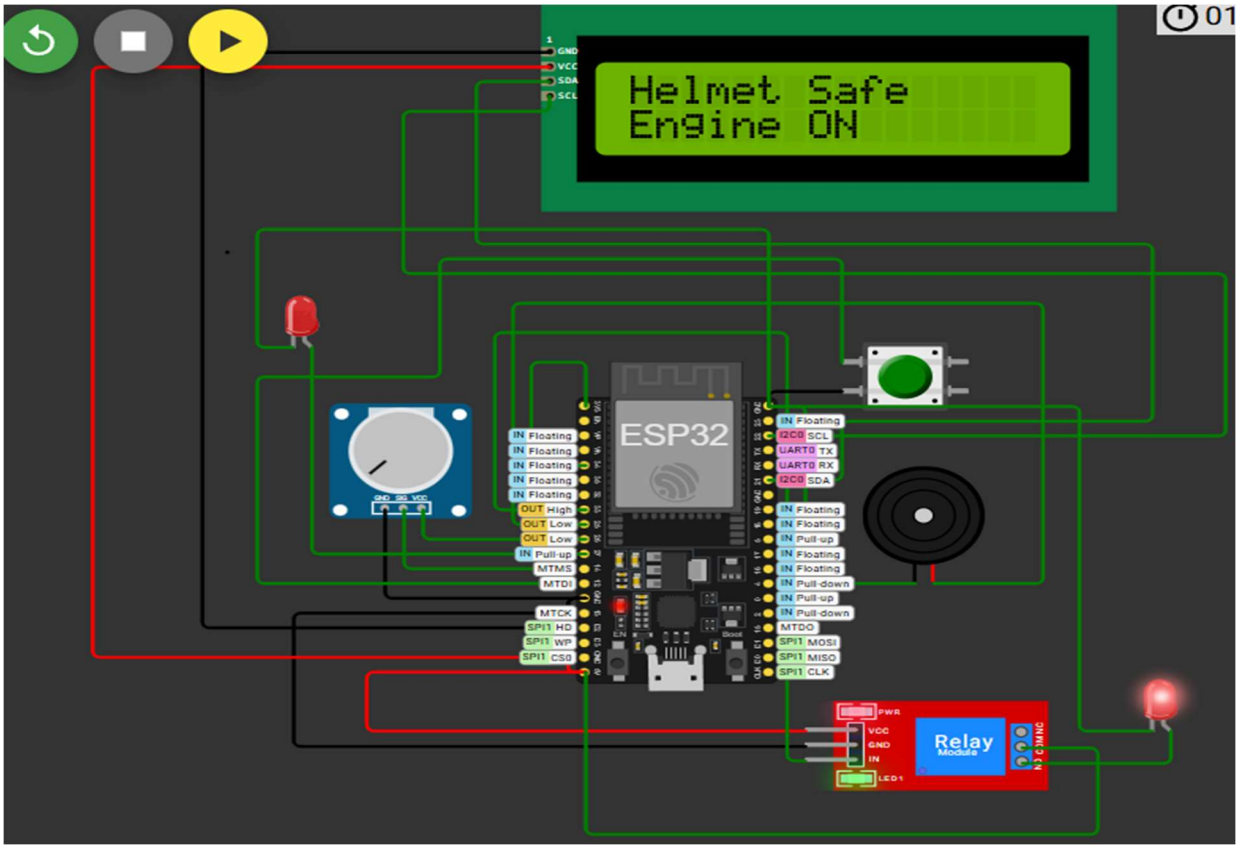
4. Software & Firmware Design

4.1 Control Logic Flow

- Initialize all sensors, GPS, GSM, and communication interfaces.
- Continuously read alcohol sensor values and compare them with the safety threshold.
- Verify helmet-wear status; if helmet is not worn or alcohol is above threshold, keep ignition locked and alert the rider locally.

- Monitor accelerometer/vibration data for sudden impacts or abnormal patterns indicating a crash.
- On accident detection, obtain GPS location and send SMS alerts to predefined contacts with coordinates or a map link.





4.2 Key Software Modules

- Sensor reading and calibration routines (MQ3, MPU6050/SW-420, IR/pressure sensor).
- Threshold configuration and decision-making logic for alcohol and accident events.
- GPS data parsing (NMEA sentences) and extraction of latitude/longitude.
- GSM communication functions for sending formatted SMS alerts.
- Ignition relay control and user feedback (buzzer, LEDs).

5. Testing & Validation

5.1 Test Scenarios

- Alcohol Detection Tests:
    - Vary alcohol concentration at the sensor and observe ignition lock and alert behavior.
- Helmet-Wear Detection Tests:
    - Check ignition permission with helmet worn vs not worn.
- Accident Simulation Tests:
    - Simulate falls or impacts to validate crash detection thresholds and alert triggering.
- GPS & GSM Tests:
    - Validate location accuracy and correctness of SMS messages (coordinates / Google Maps link).

5.2 Performance Metrics

- Alcohol detection accuracy and response time.
- False positive/negative rate for accident detection.
- Reliability and latency of GPS/GSM-based emergency alerts.

6. Implementation Phases

Phase	Description
Requirements & Research	Study existing smart helmet systems, finalize features and components.
Hardware Design	Select sensors, microcontroller, GPS/GSM, and ignition interface.
Firmware Development	Implement sensing, decision logic, GPS/GSM communication.
Integration & Testing	Integrate helmet with vehicle ignition; test all scenarios.
Optimization& Refinement	Tune thresholds, reduce false alarms, improve power usage.
Documentation	Prepare project report, diagrams, test results, and user manual.

7. Expected Outcomes

- A fully functional IoT-based smart helmet that prevents drunk riding and detects accidents automatically.
- Real-time GPS-enabled alert system that sends crash location to predefined contacts via SMS.
- Demonstrated improvement in rider safety through integrated alcohol detection, accident monitoring, and ignition control.
- Practical understanding of embedded systems, IoT communication, and safety-critical application design.

Conclusion

This project provides end-to-end experience in designing an IoT-based safety system, from sensor selection and embedded firmware development to real-time communication and system validation. By integrating alcohol detection, accident monitoring, and GPS-based emergency alerts into a single smart helmet, the

solution demonstrates how low-cost IoT technologies can significantly enhance road safety for two-wheeler riders.